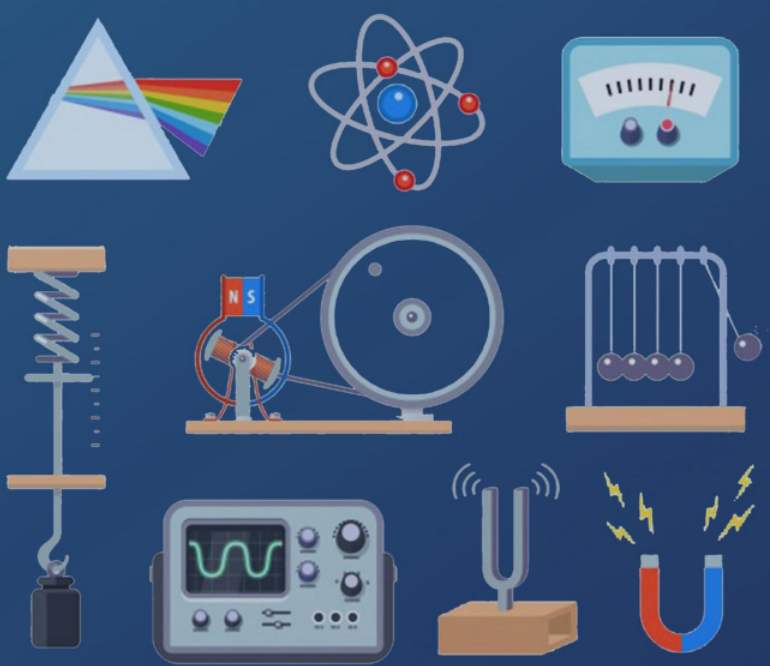
PHYSICS

Calorimetry



SHEET

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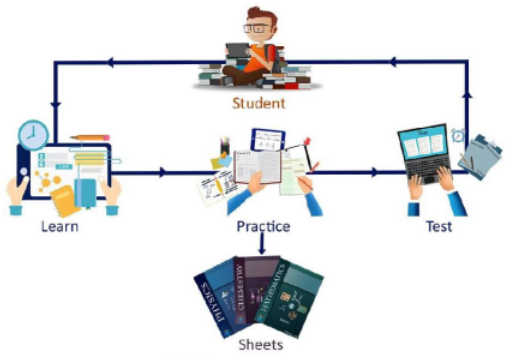
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Calorimetry	1-2	6	1,2	1
2	Principle of Calorimetry	3	1,3,5	3,4	4
3	Change of State	4-7	2,4,7	5-8	2,3,5,6,7

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Calorimetry

Q1. Heat is associated with

(A) kinetic energy of random motion of molecules.

(B) kinetic energy of orderly motion of molecules.

(C) total kinetic energy of random and orderly motion of molecules.

(D) kinetic energy of random motion in some cases and kinetic energy of orderly motion in other.

Q2. Calorie is defined as the amount of heat required to raise temperature of 1 g of water by 1°C and it is defined under which of the following conditions :–

(A) From 14.5°C to 15.5°C at 760 mm of Hg

(B) From 98.5°C to 99.5°C at 760 mm of Hg

(C) From 13.5°C to 14.5°C at 76 mm of Hg

(D) From 3.5°C to 4.5°C at 76 mm of Hg

Principle of Calorimetry

Q3. 1 kg of ice at – 10°C is mixed with 4.4 kg of water at 30°C. The final temperature of mixture is :

(specific heat of ice is 2100 J/kg/k)

(A) 2.3°C

(B) 4.4°C

(C) 5.3°C

(D) 8.7°C

Change of State

Q4. A block of mass 2.5 kg is heated to temperature of 500°C and placed on a large ice block. What is the maximum amount of ice that can melt (approx.). Specific heat for the body = 0.1 Cal/gm°C.

(A) 1 kg

(B) 1.5 kg

(C) 2 kg

(D) 2.5 kg

Q5. 10 gm of ice at 0°C is kept in a calorimeter of water equivalent 10 gm. How much heat should be supplied to the apparatus to evaporate the water thus formed? (Neglect loss of heat)

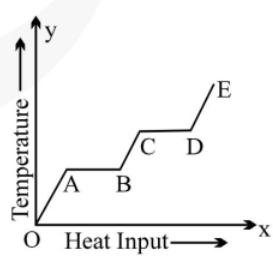
(A) 6200 cal

(B) 7200 cal

(C) 13600 cal

(D) 8200 cal

Q6. A solid material is supplied with heat at a constant rate. The temperature of material is changing with heat input as shown in the figure. What does slope DE represents?



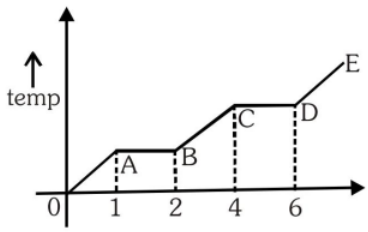
(A) latent heat of liquid

(B) latent heat of vapour

(C) heat capacity of vapour

(D) inverse of heat capacity of vapour

Q7. **Statement–1 :** A solid material is supplied heat at a constant rate. The temp. of the material is changing with the heat input as shown in figure. Latent heat of vaporization of substance is double that of fusion (given CD = 2AB).



Statement–2 : $L_f \propto AB$ and $L_v \propto CD$

(A) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. When two samples at different temperatures are mixed, the temperature of the mixture can be :-

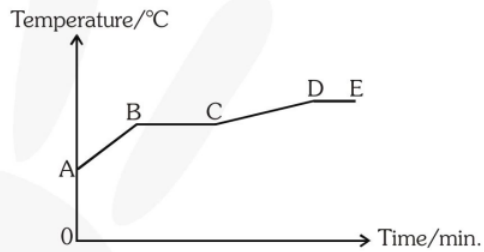
(A) lesser than lower or greater than higher temperature

(B) equal to lower or higher temperature

(C) greater than lower but lesser than higher temperature

(D) average of lower and higher temperatures.

Q2. A well insulated substance in solid state is heated at a constant rate until it vaporizes completely. The temperature-time graph of the substance is shown below. Which of the following statements is/are true ?



(A) The specific heat capacity of the substance in solid state is greater than that of the substance in liquid state.

(B) The specific latent heat of fusion of the substance is greater than the specific latent heat of vaporization of the substance.

(C) If the rate of heating increases, the slope of AB will be increased but the length of BC remains unchanged.

(D) If the mass of the substance is doubled, the length of DE is also doubled but the slope of CD is halved.

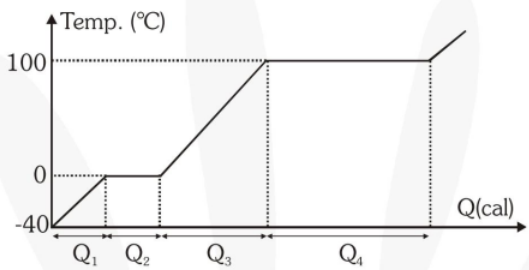
MATCH THE COLUMN TYPE

Following questions contain statements given in two columns, which have to be matched. The statements in ***Column-I*** *are labelled as A, B, C and D while the statements in* ***Column-II*** *are labelled as (P), (Q), (R) and (S). Any given statement in* ***Column-I*** *can have correct matching with* ***ONE OR MORE*** *statement(s) in* ***Column-II****.*

Q3. Three liquids A, B and C having same specific heat and mass m, 2m and 3m have temperature 20°C, 40°C and 60°C respectively. Temperature of the mixture when :

Column I	Column II
(A) A and B are mixed	(P) 35°C
(B) A and C are mixed	(Q) 52°C
(C) B and C are mixed	(R) 50°C
(D) A, B and C all three are mixed	(S) 45°C
	(T) None

Q4. Figure shows the temperature variation when heat is added continuously to a specimen of ice (10 g) at –40 °C at constant rate. (Specific heat of ice is 0.53 cal/g °C and L_{ice} = 80 cal/g, L_{water} = 540 cal/g)



Column–I	Column–II
(A) Value of Q_1 (in cal)	(P) 800
(B) Value of Q_2 (in cal)	(Q) 1000
(C) Value of Q_3 (in cal)	(R) 5400
(D) Value of Q_4 (in cal)	(S) 212
	(T) 900

(A) A→S; B→P; C→Q; D→T
(B) A→P; B→S; C→Q; D→R
(C) A→P; B→S; C→R; D→Q
(D) A→S; B→P; C→Q; D→R

SUBJECTIVE TYPE

Q5. Two identical calorimeter A and B contain equal quantity of water at 20°C. A 5 gm piece of metal X of specific heat 0.2 cal g^{-1} $(C°)^{-1}$ is dropped into A and a 5 gm piece of metal Y into B. The equilibrium temperature in A is 22°C and in B 23°C. The initial temperature of both the metals is 40°C. Find the specific heat of metal Y in cal g^{-1} $(C°)^{-1}$.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q6. Calculate heat required in cal to raise the temperature of 1 g of water through 1°C ?

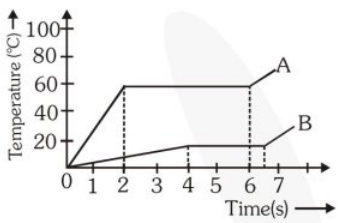
Q7. Find Heat releases by 1 kg steam at 150°C if it convert into 1 kg water at 50°C (in k cal).

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Two substances A and B of equal mass m are heated by uniform rate of 6 cal s^{-1} under similar conditions. A graph between temperature and time is shown in figure. Ratio of heat absorbed H_A / H_B by them during complete fusion is :-



(A) $\frac{9}{4}$ (B) $\frac{4}{9}$ (C) $\frac{8}{5}$ (D) $\frac{5}{8}$

Q2. The density of a material A is 1500 kg/m^3 and that of another material B is 2000 kg/m^3. It is found that the heat capacity of 8 volumes of A is equal to heat capacity of 12 volumes of B. The ratio of specific heats of A and B will be

(A) 1 : 2 (B) 3 : 1 (C) 3 : 2 (D) 2 : 1

Q3. Some steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C so that the temperature of the calorimeter and its contents rises to 80°C. What is the mass of steam condensing. (in kg)

(A) 0.130 (B) 0.065
(C) 0.260 (D) 0.135

Q4. A continuous flow water heater (geyser) has an electrical power rating = 2 kW and efficienty of conversion of electrical power into heat = 80%. If water is flowing through the device at the rate of 100 cc/sec, and the inlet temperature is 10°C, the outlet temperature will be

(A) 12.2°C (B) 13.8°C
(C) 20°C (D) 16.5°C

Q5. 540 g of ice at 0°C is mixed with 540 g of water at 80°C. The final temperature of the mixture is (Given latent heat of fusion of ice = 80 cal/g and specific heat capacity of water = 1 cal/g^0C)

(A) 0°C (B) 40°C
(C) 80°C (D) less than 0°C

Q6. 2 kg of ice at –20°C is mixed with 5 kg of water at 20°C in an insulating vessel having a negligible heat capacity. Calculate the final mass of water remaining in the container. It is given that the specific heats of water and ice are 1 kcal/kg/°C while the latent heat of fusion of ice is 80 kcal/kg :-

(A) 7 kg (B) 6 kg
(C) 4 kg (D) 2 kg

Q7. Ice at 0°C is added to 200 g of water initially at 70°C in a vacuum flask. When 50 g of ice has been added and has all melted the temperature of the flask and contents is 40°C. When a further 80g of ice has been added and has all metled, the temperature of the whole is 10°C. Calculate the specific latent heat of fusion of ice.

[Take S_w =1 cal /gm °C.]

(A) 3.8×10^5 J/ kg (B) 1.2×10^5 J/ kg
(C) 2.4×10^5 J/ kg (D) 3.0×10^5 J/ kg

Q8. In an insulated vessel, 0.05 kg steam at 373 K and 0.45 kg of ice at 253 K are mixed. Then, find the final temperature of the mixture.

Given,

L_{fusion} = 80 cal/g = 336 J/g,

$L_{vaporization}$ = 540 cal/g = 2268 J/g,

S_{ice} = 2100 J/kg K = 0.5 cal/gK and

S_{water} = 4200 J/kg K = 1 cal /gK

(A) 273 K (B) 373 K
(C) 100 K (D) 0 K

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Equal amount of heat energy are transferred into equal mass of ethyl alcohol and water sample. The rise in temperature of water sample is 25°C. The temperature rise of ethyl alcohol will be.

(Specific heat of ethyl alcohol is one half of the specific heat of water).

(A) 12.5°C

(B) 25°C

(C) 50°C

(D) It depends on the rate of energy transfer.

Q2. 50 gm ice at – 10°C is mixed with 20 gm steam at 100°C. When the mixture finally reaches its steady state inside a calorimeter of water equivalent 1.5 gm then : [Assume calorimeter was initially at 0°C, Take latent heat of vaporization of water = 540 cal/gm, Latent heat of fusion of water = 80 cal/gm, specific heat capacity of water = 1 cal/gm-°C, specific heat capacity of ice = 0.5 cal/gm°C]

(A) Mass of water remaining is : 67.4 gm

(B) Mass of steam remaining is : 2.6 gm

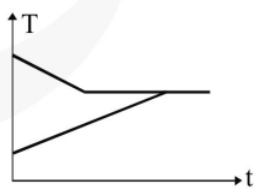
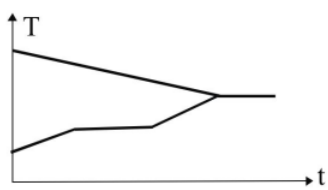
(C) Mass of water remaining is : 67.87 gm

(D) Mass of steam remaining is : 2.13 gm

MATCH THE COLUMN TYPE

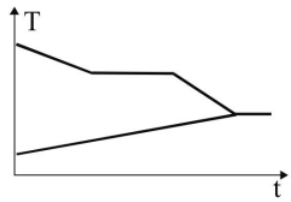
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Q3. A sample 'A' of liquid water and a sample B of ice of equal mass are kept in 2 nearby containers so that they can exchange heat with each other but are thermally insulated from the surroundings. The graphs in column-II show the sketch of temperature T of samples versus time t. Match with appropriate description in column-I.

Column I	**Column II**
(A) Equilibrium temperature is above melting point of ice.	(P) 
(B) At least some of water freezes.	(Q) 

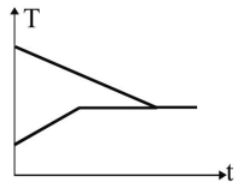
(C) At least some of ice melts.

(R)

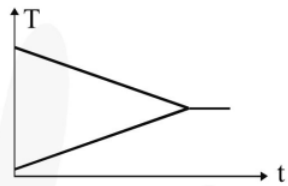


(D) Equilibrium temperature is below freezing point of water

(S)



(T)



NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q4. A flow calorimeter is used to measure the specific heat of a liquid. Heat is added at a known rate to a stream of the liquid as it passes through the calorimeter at a known rate. Then a measurement of the resulting temperature difference between the inflow and the outflow points of the liquid stream enables us to compute the specific heat of the liquid. A liquid of density 0.2 g/cm^3 flows through a calorimeter at the rate of 10 cm^3/s. Heat is added by means of a 250-W electric heating coil, and a temperature difference of 25°C is established in steady-state conditions between the inflow and the outflow points. Find the specific heat of the liquid (in J/°C kg.)

Q5. An aluminium container of mass 100 gm contains 200 gm of ice at – 20°C. Heat is added to the system at the rate of 100 cal/s. Find the temperature of the system (in °C) after 4 minutes.

(Given : specific heat of ice = 0.5 cal/gm/°C and L = 80 cal/gm, specific heat of Al = 0.2 cal/gm/°C)

SUBJECTIVE TYPE

Q6. A block of ice with mass m falls into a lake. After impact, a mass of ice $m/5$ melts. Both the block of ice and the lake have a temperature of 0°C. If L represents the latent heat of fusion, the minimum distance the ice fell before striking the surface is

Q7. 5 gm of steam at 100°C is mixed with 10gm of ice at 0°C. Find the final composition of mixture.

Answer Key

Ex-1						
1	**2**	**3**	**4**	**5**	**6**	**7**
A	A	D	B	D	D	A

Ex-1A

1. B,C,D **2.** B,D **3.** (A) T (B) R, (C) Q, (D) T **4.** D

5. 27/85 **6.** 1 cal **7.** 615 Kcal.

Ex-2							
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
C	D	A	B	A	B	A	A

Ex-2A

1. C **2.** A,B **3.** A - Q ; B - P,R ; C - Q,S ; D - R

4. 5000 J/°C kg **5.** 25.5°C **6.** $\frac{L}{5g}$ **7.** 1.67 gm of steam +13.33 gm of H_2O

-: Important Notes & Formulas :-